



BIOLOGICAL SCIENCE (BOTANY)

CLASSIFICATION OF PLANTS:

1. **BRYOPHYTES (non vascular):** plants that has no transport vessels
2. **TRACHEOPHYTES (vascular):** plants with transport vessel like xylem and phloem
 - **Gymnosperms** – cone bearing plants
 - **Angiosperms** – flowering plants
 - a) **Monocot:** flowering plants with single cotyledon or seed-leaf
 - b) **Dicot:** flowering plants with double cotyledon or seed-leaf

PLANT PARTS:

1. **Roots** – absorb water and nutrients from the soil
2. **Stem** – transport the nutrients to the leaves through cells that are specialized for transport
3. **Leaves** – food manufacturing site
 - stomata in the leaves open to exchange photosynthetic gases and close to minimize excessive water loss
 - **Photosynthesis:** means by which plants create food; requires carbon dioxide, water, chlorophyll and sunlight
 - the product of photosynthesis is glucose and oxygen which they excrete during night time
4. **Flowers** – reproductive organ of the plants which turns into a FRUIT when the embryo is pollinated

Cuttings, runners	bulbs	and	Plants which has seed-bearing fruits
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Plant Growth

- ✓ **Tropism** – directional growth or movement of a plant in reaction to stimuli.

Types of Tropism	Keyword	Definition
Electrotropism	Electro (electric)	Growth or movement in response to an electric field
Thermotropism	Thermo (heat)	Growth or movement in response to temperature
Phototropism	Photo (light)	Growth or movement towards the direction of light
Geotropism	Geo (earth)	Growth or movement towards the direction of Earth's gravity
Thigmotropism	Thigmo (touch)	Growth or movement in response to touch or contact
Hydrotropism	Hydro (water)	Growth or movement towards water or bodies of water

PLANT REPRODUCTION

Vegetative Propagation	Fertilization
Asexual reproduction	Sexual reproduction
Involves the root, stem or leaves	Involves the pollination of the flower
Produces an entirely new plant genetically identical to the parent	After fertilization, the ovule becomes the seed which later grows into a new plant, and the ripened ovary becomes the fruit
Examples are grafting,	Examples are those